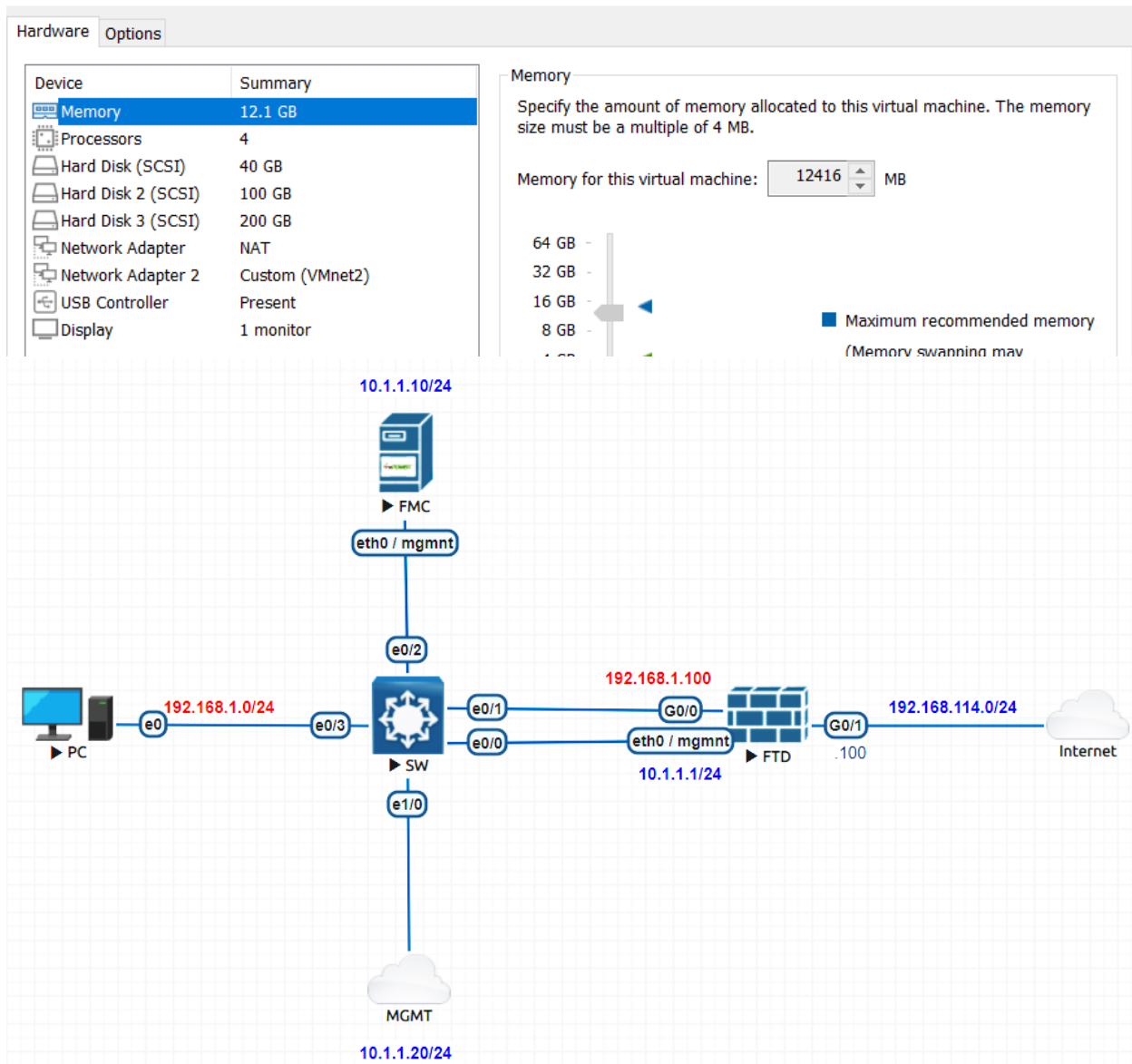


Initial Working Lab:

Lab topology using EVE NG with below Memory, Processors, Hard Disk and Interface setting.

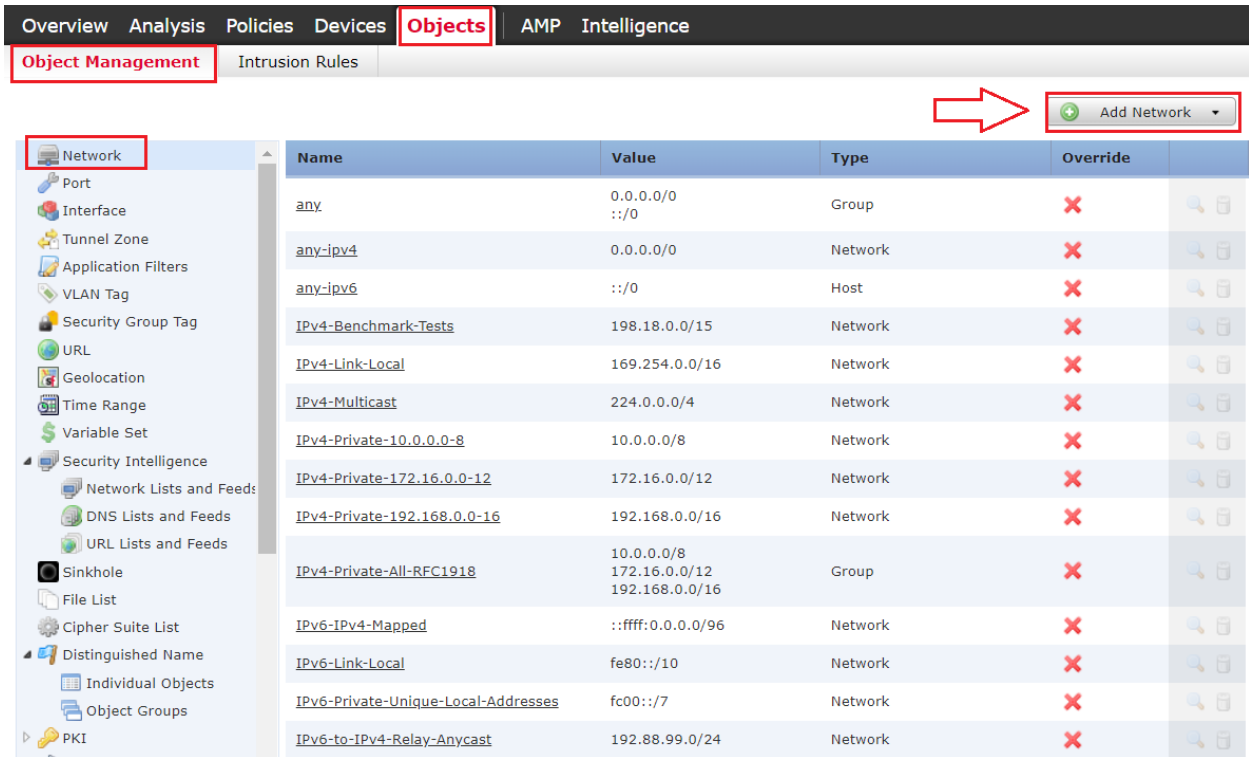
Virtual Machine Settings



FMC Management IP	10.1.1.10/24
FTD Management IP	10.1.1.1/24
FTD Internal Interface G0/0	192.168.1.100/24
FTD External Interface G0/1	192.168.114.100/24
Management Computer IP	10.1.1.20/24
Internal Test PC IP	192.168.1.1/24
ISP, Gateway IP	192.168.114.2

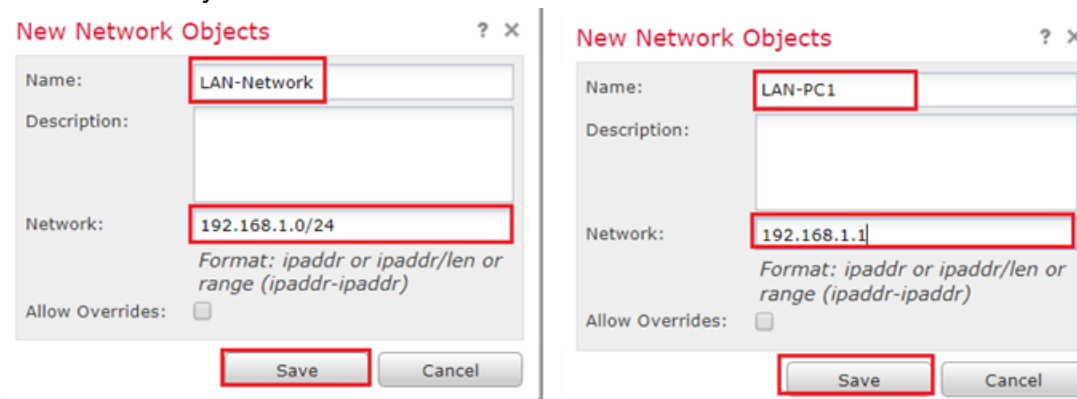
Create Objects:

Go to **Objects > Object Management > Add Network > Add Object > Add** one for you internal network(s) LAN PC, ISP IP etc Objects.



Name	Value	Type	Override
any	0.0.0.0/0 ::/0	Group	✗
any-ipv4	0.0.0.0/0	Network	✗
any-ipv6	::/0	Host	✗
IPv4-Benchmark-Tests	198.18.0.0/15	Network	✗
IPv4-Link-Local	169.254.0.0/16	Network	✗
IPv4-Multicast	224.0.0.0/4	Network	✗
IPv4-Private-10.0.0.0-8	10.0.0.0/8	Network	✗
IPv4-Private-172.16.0.0-12	172.16.0.0/12	Network	✗
IPv4-Private-192.168.0.0-16	192.168.0.0/16	Network	✗
IPv4-Private-All-RFC1918	10.0.0.0/8 172.16.0.0/12 192.168.0.0/16	Group	✗
IPv6-IPv4-Mapped	::ffff:0.0.0.0/96	Network	✗
IPv6-Link-Local	fe80::/10	Network	✗
IPv6-Private-Unique-Local-Addresses	fc00::/7	Network	✗
IPv6-to-IPv4-Relay-Anycast	192.88.99.0/24	Network	✗

Local LAN Subnet Object which is **192.168.1.0/24** and Internal test LAN PC IP address which is **192.168.1.1** Object.



New Network Objects

Name: LAN-Network

Description:

Network: 192.168.1.0/24
Format: ipaddr or ipaddr/len or range (ipaddr-ipaddr)

Allow Overrides: ☐

Save Cancel

New Network Objects

Name: LAN-PC1

Description:

Network: 192.168.1.1
Format: ipaddr or ipaddr/len or range (ipaddr-ipaddr)

Allow Overrides: ☐

Save Cancel

Create Security Zone:

Go to **Objects > Object Management**. Choose **Interface** from the list of object types. Click **Add > Security Zone**. Enter a Name. Choose an Interface Type. From the **Device > Interfaces** drop-down list, choose a device that contains interfaces you want to add. Click **Save**.

The screenshot shows the 'Security Zones' configuration window. The 'Name' field is set to 'Outside' and 'Interface Type' is set to 'Routed'. Under 'Available Interfaces', 'FTD-1' is selected. The 'Add' button is visible between the two interface lists. The 'Save' button is highlighted at the bottom right.

Finally created two Security Zones, **Inside** and **Outside**

The screenshot shows the 'Object Management' page with the 'Objects' tab selected. The 'Interface' object type is selected. A table lists the created security zones: 'Inside' and 'Outside', both of type 'Security Zone' and 'Routed' interface type. The 'Add' button is highlighted at the top right.

Name	Type	Interface Type
Inside	Security Zone	Routed
Outside	Security Zone	Routed

Configure FTD Interfaces:

Select **Devices > Device Management** and click the edit icon **pencil** for your Firepower Threat Defense device. Click the edit icon (**pencil**) for the interface you want to edit. In the Mode drop-down list, choose **None**. For Routed interface the Mode is: **None**, Enable the interface by checking the **Enabled** check box. Give Name to Interface, Name is equivalent to the ASA interface name if On FTD all interfaces have security level = 0, Choose Security Zone, Configure IPV4 address in our case **192.168.114.100/24**

The screenshot shows the Cisco Firepower Threat Defense (FTD) configuration interface. The top navigation bar includes 'Overview', 'Analysis', 'Policies', 'Devices', 'Objects', 'AMP', and 'Intelligence'. The 'Devices' tab is selected, and the 'Device Management' sub-tab is active. The main panel displays 'FTD-1' and 'Cisco Firepower Threat Defense for KVM'. The 'Interfaces' tab is selected, showing a list of interfaces: GigabitEthernet0/0, GigabitEthernet0/1, GigabitEthernet0/2, GigabitEthernet0/3, and Diagnostic0/0. The 'GigabitEthernet0/1' interface is highlighted. The 'Edit Physical Interface' dialog is open, showing the following configuration:

- Mode: None
- Name: Outside
- Enabled: ☒ (Management Only: ☐)
- Security Zone: Outside
- Description: Connected to WAN
- General tab: IPv4
- IP Type: Use Static IP
- IP Address: 192.168.114.100/24

Red boxes highlight the 'Mode' dropdown, the 'Enabled' checkbox, the 'Name' field, the 'Security Zone' dropdown, the 'IPv4' tab, and the 'IP Address' field. A red arrow points to the 'Enabled' checkbox. The 'OK' and 'Cancel' buttons are at the bottom right.

Same way configures second interface for inside zone

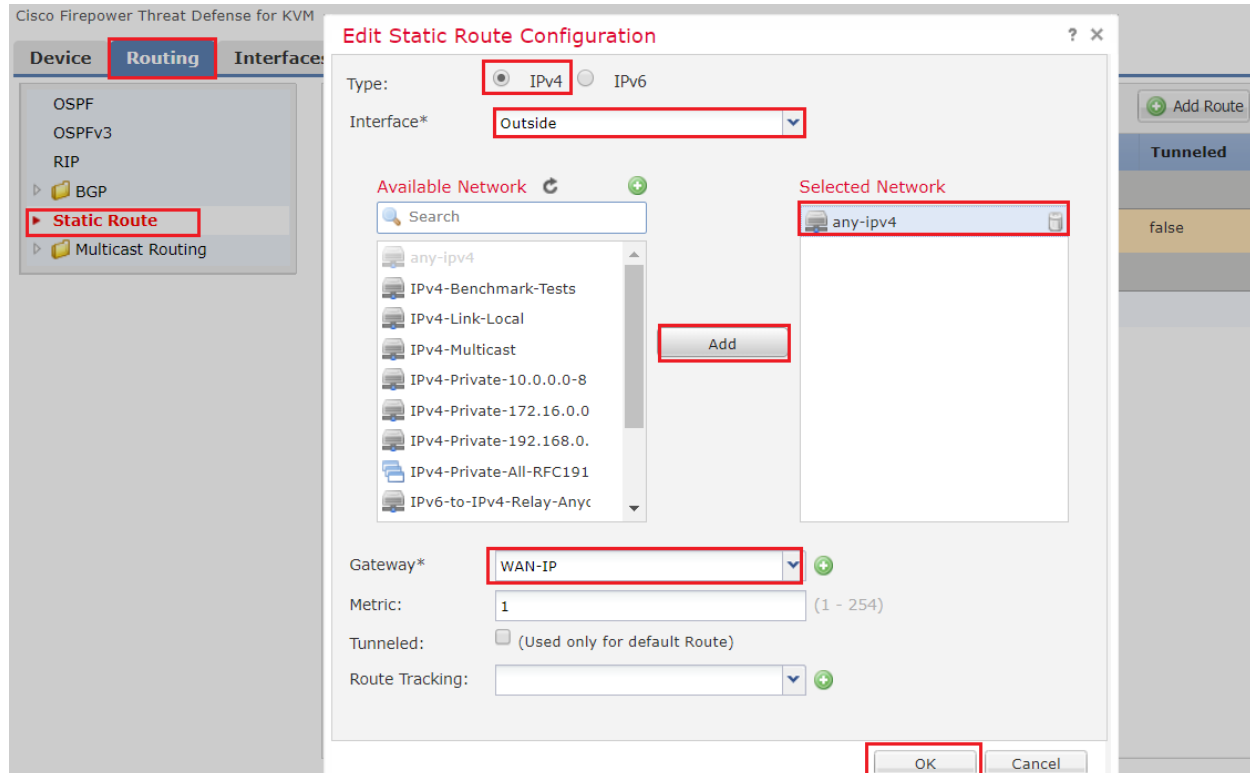
The screenshot shows the Cisco Firepower Threat Defense (FTD) configuration interface. The top navigation bar includes 'Overview', 'Analysis', 'Policies', 'Devices', 'Objects', 'AMP', and 'Intelligence'. The 'Devices' tab is selected, and the 'Device Management' sub-tab is active. The main panel displays 'FTD-1' and 'Cisco Firepower Threat Defense for KVM'. The 'Interfaces' tab is selected, showing a list of interfaces: GigabitEthernet0/0, GigabitEthernet0/1, GigabitEthernet0/2, GigabitEthernet0/3, and Diagnostic0/0. The 'GigabitEthernet0/0' interface is highlighted. The 'Edit Physical Interface' dialog is open, showing the following configuration:

- Mode: None
- Name: Inside
- Enabled: ☒ (Management Only: ☐)
- Security Zone: Inside
- Description: Connected to LAN
- General tab: IPv4
- IP Type: Use Static IP
- IP Address: 192.168.1.100/24

Red boxes highlight the 'Mode' dropdown, the 'Enabled' checkbox, the 'Name' field, the 'Security Zone' dropdown, the 'IPv4' tab, and the 'IP Address' field. A red arrow points to the 'Enabled' checkbox. The 'OK' and 'Cancel' buttons are at the bottom right.

Configure Static Route:

Choose **Devices > Device Management** and edit the Firepower Threat Defense device. Click the **Routing** tab. Select **Static Route** from the table of contents. Click **Add Routes**. Click the **IPv4** radio button depending on the type of static route that you are adding. Choose the Interface to which this static route applies in this case Outside interface. In the Available Network list, choose the destination network. To define a default route, create an object with the address 0.0.0.0/0 and select it here. In the **Gateway** field, enter or choose the gateway router which is the next hop for this route. You can provide an IP address or a Networks/Hosts object. Click **Ok**.

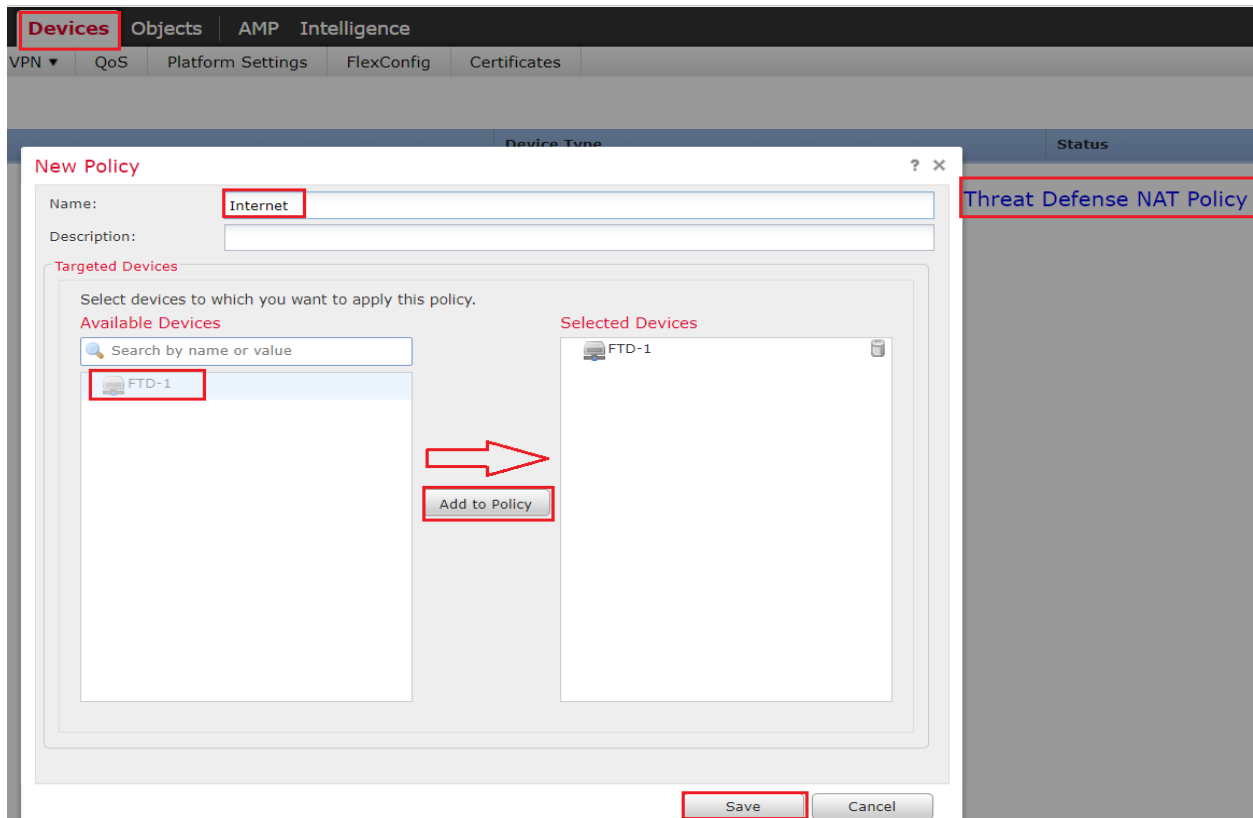


Finally, Default route is look like below any destination send to WAN-IP which is Internet Service Provider IP 192.168.114.2.

Network	Interface	Gateway	Tunneled	Metric
any-ipv4	Outside	WAN-IP	false	1

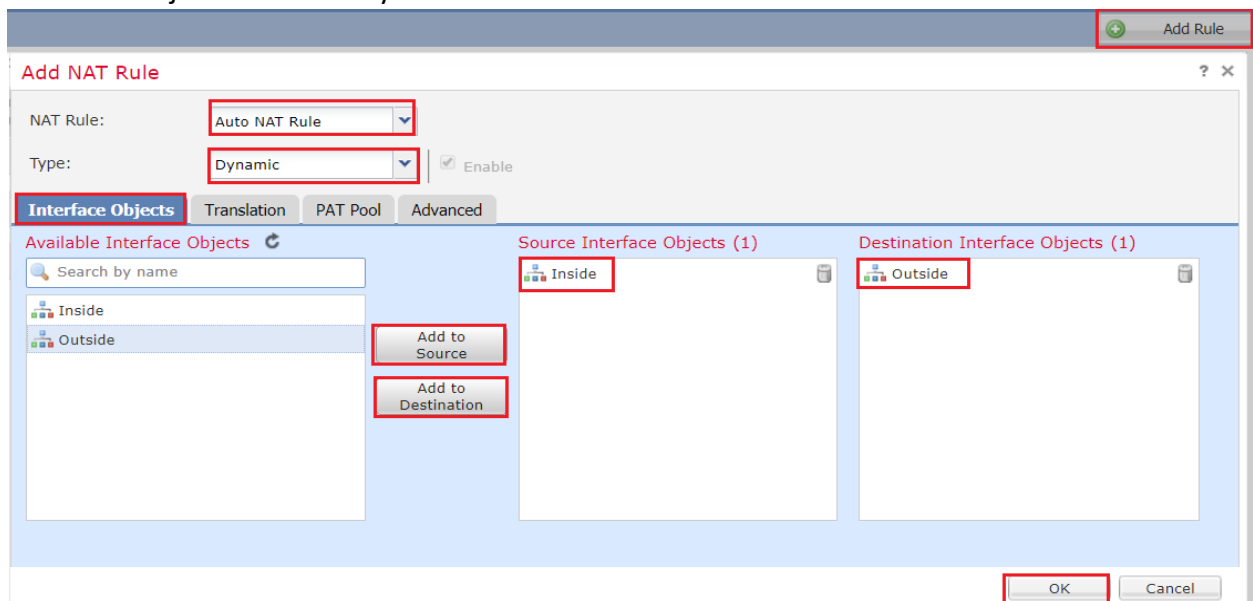
Configure Network Address Translation (NAT):

Select **Devices > NAT**. Click New Policy > Threat Defense NAT to create a new policy. Give the policy a name, optionally assign devices to it, and click **Save**.



Click the **Add Rule** button to create a new rule. NAT Rule—Select **Auto NAT Rule**.

Type—Select **Dynamic**. On the **Interface Objects** tab, Source Interface Objects, Destination Interface Objects like security zones.



On the Translation tab, configure Original Source—The network object that contains the addresses you are translating. Translated Source—The network object or group that contains the mapped addresses. Click **Save** to add the rule.

Add NAT Rule

NAT Rule: Auto NAT Rule

Type: Dynamic ☒ Enable

Interface Objects **Translation** PAT Pool Advanced

Original Packet

Original Source:* LAN-Network

Original Port: TCP

Translated Packet

Translated Source: Destination Interface IP

Translated Port:

OK Cancel

Click **Save** on the NAT page to save your changes and click **Deploy** to apply the changes to FTD.

Deploy Policies Version: 2020-11-01 09:28 AM

Device	Inspect Interruption	Type	Group	Current Version
FTD-1	No	FTD		2020-11-01 12:58 AM

Please Wait

Initiating deploy process...

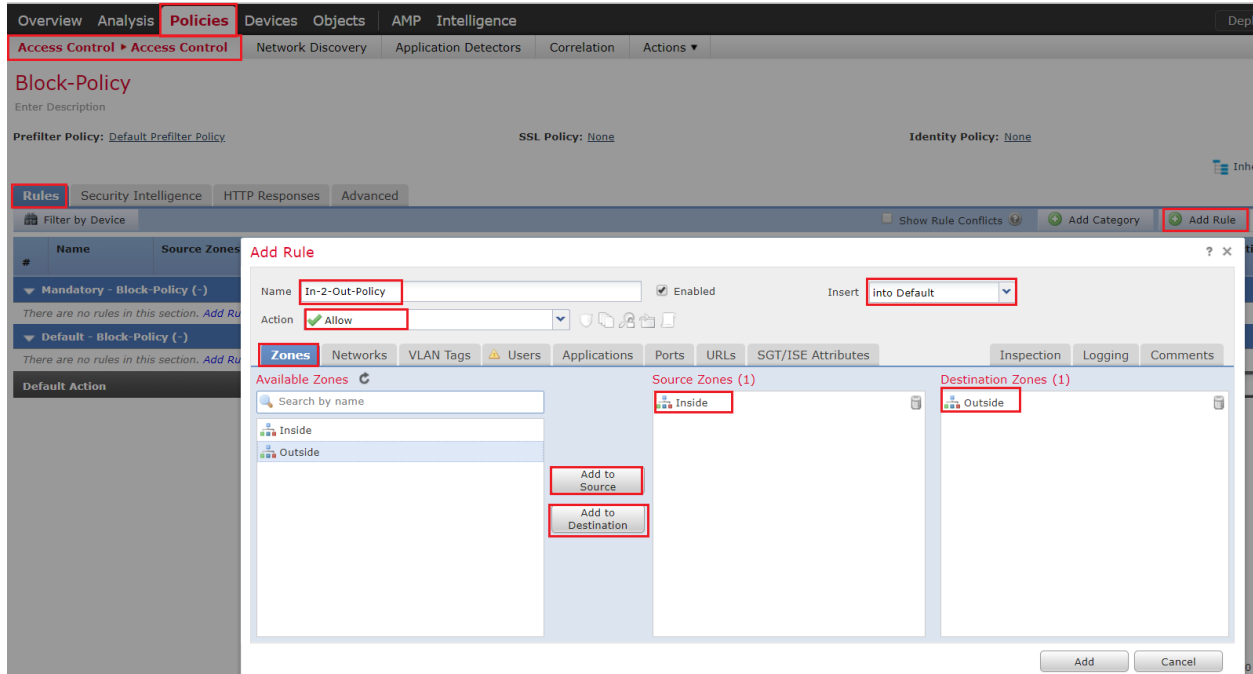
Selected devices: 1

Deploy Cancel

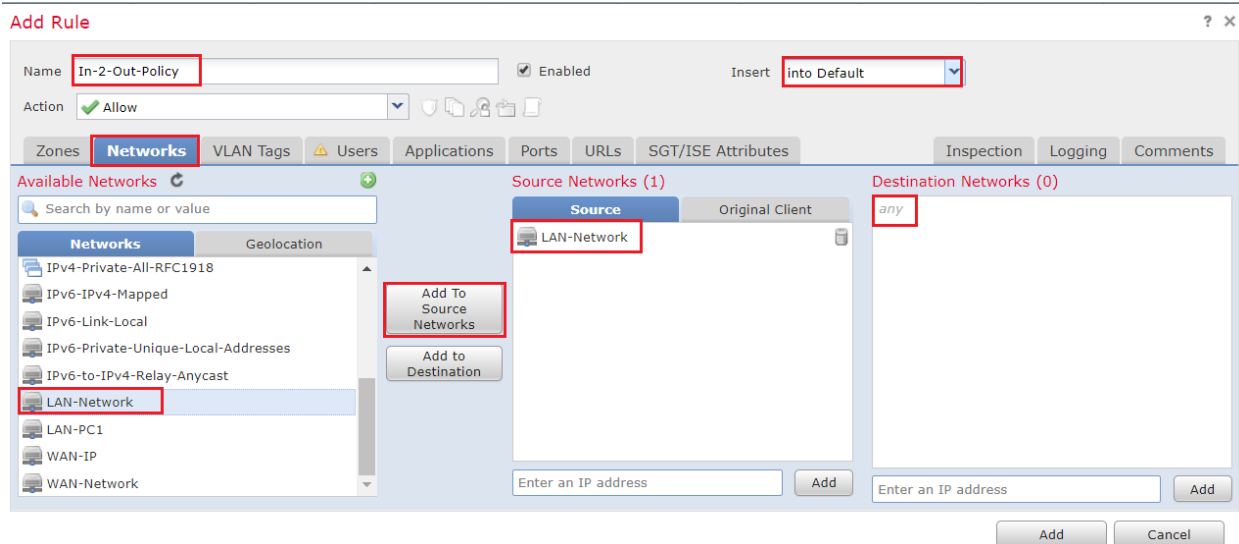
Save Cancel

Create Policy:

Choose **Policies > Access Control**, Click New **Policy**. Enter a unique Name. To add a new rule, click Add Rule. Enter a Name. Enabled—Specify whether the rule is Enabled. Action—Choose a rule Action. In Zones tab select Source Zone and Destination Zone.



In the **Networks** Tab select **Source Networks** and set **Destination Networks** to any.



In **Inspection** tab from Intrusion Policy select **Security Over Connectivity**.

Add Rule

Name: In-2-Out-Policy ☒ Enabled Insert: into Default

Action: ☒ Allow

Zones Networks VLAN Tags Users Applications Ports URLs SGT/ISE Attributes **Inspection** Logging Comments

Intrusion Policy: **Security Over Connectivity** Variable Set: Default Set

File Policy: None

Add Cancel

In **Logging** Tab enable **Log at Beginning of Connection** and Click **Add**.

Add Rule

Name: In-2-Out-Policy ☒ Enabled Insert: into Default

Action: ☒ Allow

Zones Networks VLAN Tags Users Applications Ports URLs SGT/ISE Attributes Inspection **Logging** Comments

☒ Log at Beginning of Connection

☐ Log at End of Connection

File Events:

☐ Log Files

Send Connection Events to:

☒ Event Viewer

☐ Syslog Select a Syslog Alert Configuration...

☐ SNMP Trap Select an SNMP Alert Configuration...

Add Cancel

Last Save the setting by clicking Save button and Deploy to FTD.

Deploy Policies Version: 2020-11-01 10:10 AM

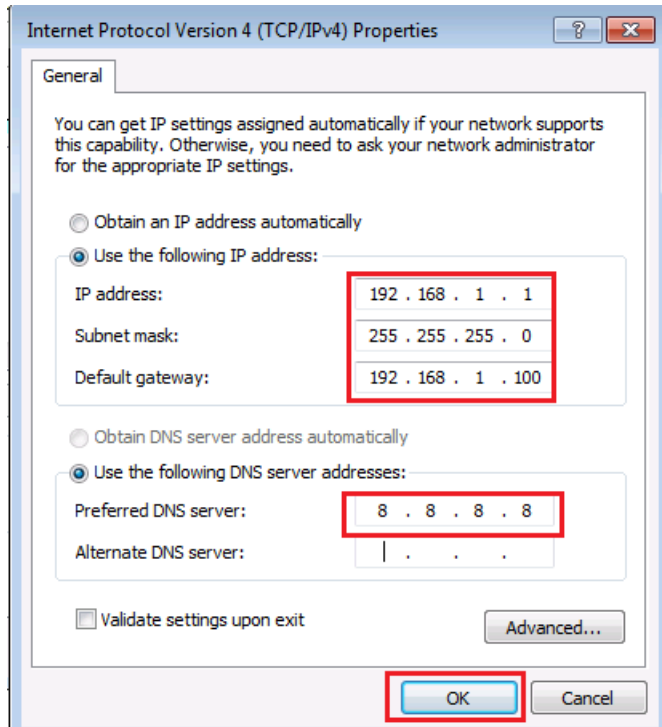
Device	Inspect Interruption	Type	Group	Current Version
☒ FTD-1	No	FTD		2020-11-01 09:28 AM

Selected devices: 1

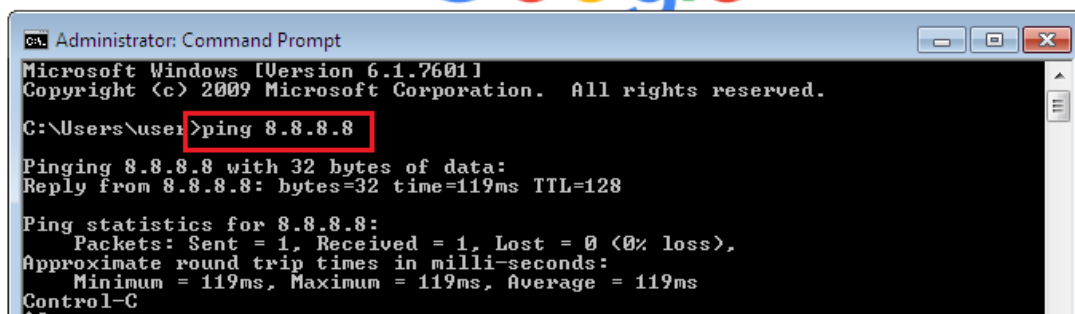
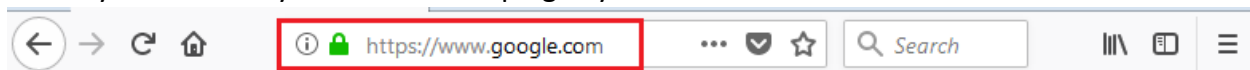
Deploy Cancel

Testing and Verification:

Let's configure Internal LAN PC given IP is 192.168.1.1, gateway and DNS details.



Let's try to access any website also to ping any outside Internet IP in this case 8.8.8.8.



Verify Events and Logs:

Go to **Analysis > Connections > Events** to verify and check Internal LAN PC events.

Overview

Analysis

Policies

Devices

Objects

AMP

Intelligence

Deploy

Context Explorer

Connections > Events

Intrusions

Files

Hosts

Users

Vulnerabilities

Correlation

Custom

Lookup

Search

Connection Events

(switch workflow)

Connections with Application Details

> Table View of Connection Events

No Search Constraints

(Edit Search)

Jump to...

	First Packet	Last Packet	Action	Reason	Initiator IP	Initiator Country	Responder IP	Responder Country	Ingress Security Zone	Egress Security Zone	Source Port / ICMP Type	Destination Port / ICMP Code	Application Protocol
↓	2020-11-01 10:38:33		Allow		192.168.1.1		8.8.8.8	USA	Inside	Outside	58348 / udp	53 (domain) / udp	DNS
↓	2020-11-01 10:38:23		Allow		192.168.1.1		8.8.8.8	USA	Inside	Outside	62406 / udp	53 (domain) / udp	DNS
↓	2020-11-01 10:38:23		Allow		192.168.1.1		8.8.8.8	USA	Inside	Outside	60846 / udp	53 (domain) / udp	DNS
↓	2020-11-01 10:38:23		Allow		192.168.1.1		143.204.215.95	USA	Inside	Outside	49230 / tcp	443 (https) / tcp	
↓	2020-11-01 10:38:23		Allow		192.168.1.1		8.8.8.8	USA	Inside	Outside	49164 / udp	53 (domain) / udp	DNS
↓	2020-11-01 10:38:21		Allow		192.168.1.1		8.8.8.8	USA	Inside	Outside	53643 / udp	53 (domain) / udp	DNS
↓	2020-11-01 10:38:21		Allow		192.168.1.1		8.8.8.8	USA	Inside	Outside	49867 / udp	53 (domain) / udp	DNS
↓	2020-11-01 10:38:00		Allow		192.168.1.1		8.8.8.8	USA	Inside	Outside	61134 / udp	53 (domain) / udp	DNS
↓	2020-11-01 10:37:29		Allow		192.168.1.1		8.8.8.8	USA	Inside	Outside	62998 / udp	53 (domain) / udp	DNS
↓	2020-11-01 10:37:29		Allow		192.168.1.1		8.8.8.8	USA	Inside	Outside	58547 / udp	53 (domain) / udp	DNS
↓	2020-11-01 10:37:29		Allow		192.168.1.1		8.8.8.8	USA	Inside	Outside	60121 / udp	53 (domain) / udp	DNS
↓	2020-11-02 14:56:55		Allow		192.168.1.1		8.8.8.8	USA	Inside	Outside	58351 / udp	53 (domain) / udp	DNS
↓	2020-11-02 14:55:51		Allow		192.168.1.1		8.8.8.8	USA	Inside	Outside	58343 / udp	53 (domain) / udp	DNS
↓	2020-11-02 14:54:46		Allow		192.168.1.1		8.8.8.8	USA	Inside	Outside	58335 / udp	53 (domain) / udp	DNS
↓	2020-11-02 14:53:43		Allow		192.168.1.1		8.8.8.8	USA	Inside	Outside	58327 / udp	53 (domain) / udp	DNS
↓	2020-11-02 14:52:40		Allow		192.168.1.1		8.8.8.8	USA	Inside	Outside	58319 / udp	53 (domain) / udp	DNS
↓	2020-11-02 14:51:36		Allow		192.168.1.1		8.8.8.8	USA	Inside	Outside	53665 / udp	53 (domain) / udp	DNS
↓	2020-11-02 14:50:34		Allow		192.168.1.1		8.8.8.8	USA	Inside	Outside	53657 / udp	53 (domain) / udp	DNS
↓	2020-11-02 14:50:34		Allow		192.168.1.1		172.217.18.46	USA	Inside	Outside	49165 / tcp	443 (https) / tcp	
↓	2020-11-02 14:50:33		Allow		192.168.1.1		8.8.8.8	USA	Inside	Outside	64415 / udp	53 (domain) / udp	DNS
↓	2020-11-02 14:50:33		Allow		192.168.1.1		8.8.8.8	USA	Inside	Outside	58159 / udp	53 (domain) / udp	DNS
↓	2020-11-02 14:49:21		Allow		192.168.1.1		8.8.8.8	USA	Inside	Outside	59917 / udp	53 (domain) / udp	DNS
↓	2020-11-02 14:49:21		Allow		192.168.1.1		8.8.8.8	USA	Inside	Outside	55899 / udp	53 (domain) / udp	DNS
↓	2020-11-02 14:49:21		Allow		192.168.1.1		216.58.212.110	USA	Inside	Outside	49161 / tcp	443 (https) / tcp	

Overview	Analysis	Policies	Devices	Objects	AMP	Intelligence
Dashboards	Reporting	Summary				

Summary Dashboard

Provides a summary of activity on the appliance

